

BRAC UNIVERSITY
Department of Computer Science and Engineering

Quiz 04
Semester: Fall 2024

Time: 25 min
Full Marks: 20

Name:	ID:	Section:
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CSE 251: Electronic Devices and Circuits

1. [CO1] 2+3
 - a) In the S model of a BJT, **identify** the mode of operation when the BJT is OFF and the mode of operation when it is ON.
 - b) **Show** the I-V characteristics graph of BJT (I_c vs V_{CE}) . **Identify** the operating regions in the graph.
2. [CO3] Implement the logic function, $f = \overline{(A + B)\overline{C}.D}$ using BJTs. 5

3. [CO2]

10

Analyze the following circuit to determine all node voltages and branch currents. Assume $\beta = 100$. You must validate your assumptions.

